

## Project Design Phase-II

### Solution Requirements (Functional & Non-functional)

|                      |                                                                |
|----------------------|----------------------------------------------------------------|
| <b>Date</b>          | 30 June 2026                                                   |
| <b>Name</b>          | Nadin Surya S                                                  |
| <b>Project Name</b>  | Meeting Room Booking System using ServiceNow App Engine Studio |
| <b>Maximum Marks</b> | 4 Marks                                                        |

#### Functional Requirements

Following are the functional requirements of the proposed Meeting Room Booking System solution.

| FR No. | Functional Requirement (Epic)      | Sub Requirement (Story / Sub-Task)                                                                                                                                                                                                                     |
|--------|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-1   | User Authentication & Access       | <ul style="list-style-type: none"> <li>• Login to the application via ServiceNow portal credentials (SSO)</li> <li>• Role-based access control for Employee and Administrator</li> </ul>                                                               |
| FR-2   | Meeting Room Booking               | <ul style="list-style-type: none"> <li>• Submit booking request via online form</li> <li>• Select Room Name, Booking Date, Start Time and End Time</li> <li>• Enter Employee Name, Employee ID, Employee Email and Purpose</li> </ul>                  |
| FR-3   | Booking Validation                 | <ul style="list-style-type: none"> <li>• Client-side validation that End Time is greater than Start Time</li> <li>• Mandatory field validation before submission</li> </ul>                                                                            |
| FR-4   | Booking Status Management          | <ul style="list-style-type: none"> <li>• Auto-assign default status as Pending on creation (Business Rule)</li> <li>• Administrator can change status to Approved</li> <li>• Administrator can change status to Completed after the meeting</li> </ul> |
| FR-5   | Workflow Automation & Notification | <ul style="list-style-type: none"> <li>• Flow Designer triggers on new booking creation</li> <li>• Automated email notification sent to employee upon booking</li> </ul>                                                                               |
| FR-6   | Reporting & Dashboard              | <ul style="list-style-type: none"> <li>• Dashboard with total bookings, status chart, room-wise chart and trend chart</li> <li>• Reports for booking status, room-wise usage, trend and booking list</li> </ul>                                        |

#### Non-Functional Requirements

Following are the non-functional requirements of the proposed Meeting Room Booking System solution.

| NFR No. | Non-Functional Requirement | Description                                                                                                                                                       |
|---------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NFR-1   | Usability                  | The application provides a simple, intuitive form-based interface so employees can book a room in a few clicks without training.                                  |
| NFR-2   | Security                   | Access is restricted via ServiceNow role-based access control; only authenticated users can create bookings and only administrators can approve or complete them. |
| NFR-3   | Reliability                | The system must reliably save every booking record and consistently apply validation and status rules without data loss.                                          |

| NFR No. | Non-Functional Requirement | Description                                                                                                                                            |
|---------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| NFR-4   | Performance                | Booking form submission, validation and status updates should complete within 2–3 seconds under normal load.                                           |
| NFR-5   | Availability               | The application should be available during business hours with minimal downtime, leveraging ServiceNow's standard platform uptime.                     |
| NFR-6   | Scalability                | The system should support an increasing number of employees, rooms and bookings without degradation in performance.                                    |
| NFR-7   | Maintainability            | Business Rules, Client Scripts and Flows should be modular and well-documented so the application can be easily extended (e.g. new rooms, new fields). |